AAPL -- ANN Stock Price Prediction Report

Generated: 2026-04-19 19:48:11 | Framework: TensorFlow/Keras | Data: yfinance

Trained for 270 epochs | Best validation loss: 0.019046

EVALUATION METRICS

MAPE (%)	RMSE	MAE
3.5037	10.7449	8.8403

R2 SCORE	DIRECTIONAL ACCURACY (%)	MAX SINGLE ERROR
0.5687	46.33	24.9836

WITHIN 5% ACCURACY (%)
70.79

RESULT INTERPRETATION

MAPE (Mean Absolute Percentage Error): How far off predictions are on average, as a percentage. Under 5% is considered good for stock prediction.

RMSE (Root Mean Squared Error): Average error in the same unit as price (e.g. USD). Larger errors are penalized more.

R2 Score: Measures how well the model explains price variance. 1.0 = perfect, 0.0 = no better than predicting the mean price.

Directional Accuracy: What percentage of UP/DOWN movements the ANN called correctly. Above 55% is meaningful -- random guessing gives 50%.

Within 5% Accuracy: Percentage of predictions that landed within 5% of the actual price.

LIVE PREDICTION SUMMARY

METRIC	VALUE
Last Close Price	270.23
ANN Predicted Price	248.9894
Expected Change	-7.860%
Direction Signal	DOWN
Prediction Time	2026-04-19T19:48:11

ARCHITECTURE AND ANALYSIS

TOPIC	DETAILS
Architecture	3 hidden layers (128 -> 64 -> 32) with BatchNorm + Dropout
Why ANN	Captures non-linear patterns that linear models miss
Training	270 epochs with early stopping + ReduceLROnPlateau
Loss Function	Huber Loss (robust to price outliers/spikes)
Optimizer	Adam with learning rate decay
Model Grade	GOOD (< 5% error)
Limitation 1	Cannot predict news-driven sudden moves
Limitation 2	Requires periodic retraining as market changes

ANN HYPERPARAMETERS

PARAMETER	VALUE
hidden_layers	[64, 32, 16]
activation	relu

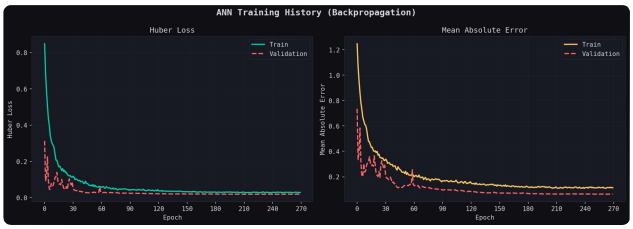
output_activation	**linear**
dropout_rate	**0.2**
l2_lambda	**0.0001**
learning_rate	**0.0005**
batch_size	**32**
epochs	**300**
patience	**20**
reduce_lr_patience	**10**
reduce_lr_factor	**0.5**
min_lr	**1e-06**

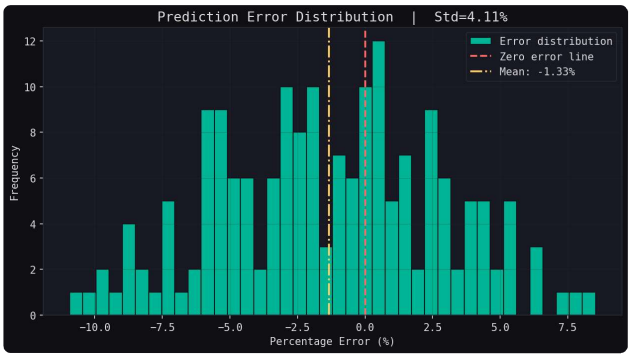
LIMITATIONS

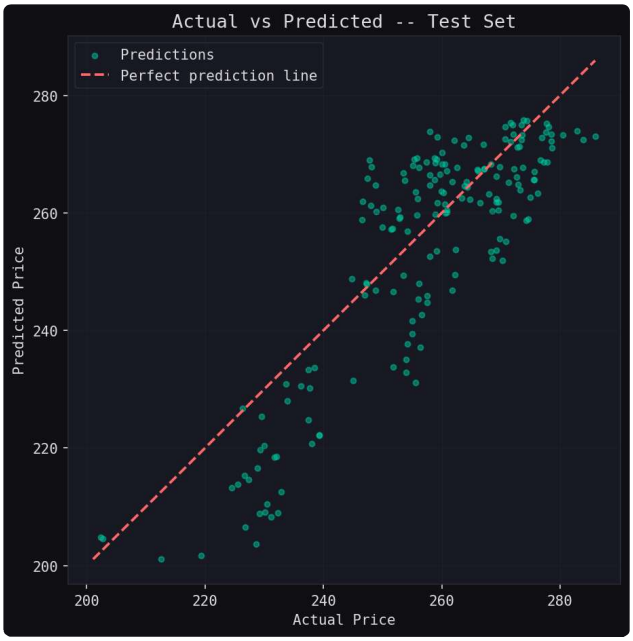
> 1. ANN cannot predict sudden price shocks caused by news, earnings surprises, or geopolitical events -- these are not present in historical price data.
> 2. The model learns from past patterns. When market conditions change significantly (e.g. from bull to bear market), the model should be retrained.
> 3. Predictions are for the next trading day. Multi-day or multi-week forecasts will have increasing error.
> 4. Volume spikes or circuit breakers can break the model's assumptions.

VISUALIZATIONS

AAPL -- Actual vs ANN Predicted Price (Test Set)
Actual Price
ANN Predicted
Prediction Gap
Price
Residual
Date
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
2026-02
2026-03
2026-04


ANN Training History (Backpropagation)
Huber Loss
Mean Absolute Error
Train
Validation
Epoch


Prediction Error Distribution | Std=4.11%
Error distribution
Zero error line
Mean: -1.33%
Frequency
Percentage Error (%)


Actual vs Predicted -- Test Set
Predictions
Perfect prediction line
Predicted Price
Actual Price

Stock Market Price Prediction Using Artificial Neural Network (ANN) on Live Data from yfinance | Major Project | Python + TensorFlow/Keras + yfinance + scikit-learn + Matplotlib